Unit: 6 Unit 7: Factoring
Topic: Difference of Squares

Name: ______________
Date: __________

Advanced (Habanero)

Q1. Factor the expression $x^2 - 4$
(A) $(x+2)(x-2)$
(B) $x(x-4)$
(C) $x(x+4)$
(D) $2(x-2)$
(E) $2(x+2)$

Q2. Factor $9y^2 - 25$
(A) $(3y+5)(3y-5)$
(B) $(3y-5)(3y+5)$
(C) $(9y+25)(y-1)$
(D) $(9y-25)(y+1)$
(E) $y(9y-25)$

Q3. Factor $a^2 - b^2$
(A) $(a+b)(a-b)$
(B) $(a-b)(a+b)$
(C) $a(a-b)$
(D) $a(a+b)$
(E) $b(a+b)$

Q4. What is the factored form of $64 - 36$
(A) $(8+6)(8-6)$
(B) $(8-6)(8+6)$
(C) $2(32-18)$
(D) $4(16-9)$
(E) $2(8-6)$

Q5. Solve for x in $x^2 - 9 = 16$ using difference of squares
(A) $x = 5$ or $x = -5$
(B) $x = 7$ or $x = -7$
(C) $x = 3$ or $x = -3$
(D) $x = 4$ or $x = -4$
(E) $x = 2$ or $x = -2$

Q6. Factor the expression $36y^2 - 81$
(A) $9(4y^2 - 9)$
(B) $9(4y^2 + 9)$
(C) $9(2y+3)(2y-3)$
(D) $9(2y-3)(2y+3)$
(E) $6(6y^2 - 9)$

Q7. What is the factored form of $16x^2 - 49y^2$
(A) $(4x+7y)(4x-7y)$
(B) $(4x-7y)(4x+7y)$
(C) $x(16x-49y)$
(D) $y(16x-49y)$
(E) $16x^2 + 49y^2$

Q8. Which of the expressions is in the form of $a^2 - b^2$
(A) $9x^2 - 4y^2 + 12$
(B) $3x^2 + 12y^2$
(C) $4x^2 + 9y^2$
(D) $x^2 + y^2$
(E) $9x^2 - 4y^2$

Open Ended Questions (FRQ)

Q9. Factor the expression $25x^2 - 16$ and show all steps

Q10. Factor $121y^2 - 144$ and identify the values of a and b

Chef's Tips

- Tip 1: Ask students to show their work and explain their reasoning
- Tip 2: Check that students are factoring correctly using the difference of squares formula
- Tip 3: Encourage students to check their answers by multiplying the factors